



Trivibhuwan University

Faculty of Humanities and Social Science

Online Inventory System

PROJECT REPORT

Submitted To:

Department of Computer Application Arunima College

A partially Fulfillment of the requirements for Bachelor of Computer Application

Submitted By:

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&

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Arunima College

Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **Subodh Raj Karki** and **Prajwol Sainju** entitled “**Online Inventory System**” in partial fulfillment of the requirements for the degree of **Bachelor of Computer Application** is recommended for the final evaluation.

.....

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LETTER OF APPROVAL

This is to certify that this project prepared by **Subodh Raj Karki** and **Prajwol Sainju** entitled “**Online Inventory System**” in partial fulfillment of the requirements for the degree of **Bachelor in Computer Application** has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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SIGNATURE of Internal Examiner Internal Examiner 	SIGNATURE of External Examiner External Examiner

ABSTRACT

This project is a web-based inventory management system built using PHP, MySQL, HTML, CSS, and JavaScript. The objective is to provide businesses with an efficient system to manage stock, track sales and purchases, and maintain product records in a digital format.

The system enables users to add, edit, and manage products, track stock levels, and view transaction history with ease. It also includes user authentication to ensure secure access. Data validation and error handling mechanisms are incorporated to prevent incorrect entries and enhance reliability.

This document provides an overview of the system, including its objectives, methodology, and expected deliverables.

Keywords: Inventory, Database, PHP, MySQL, Stock Management, Transactions, User Authentication.

ACKNOWLEDGMENT

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We are also grateful to my classmates and friends for offering me advice and moral support. To my family, thank you for encouraging me in all of my pursuits and inspiring us to follow my dreams. We are especially grateful to my parents, who supported me emotionally, believed in us and wanted the best for us.

Subodh Raj Karki & Prajwol Sainju

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LIST OF ABBREVIATIONS

Abbreviation	Definition
OIS	Online Inventory System
BCA	Bachelor in Computer Application
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
PHP	Hypertext Preprocessor
DFD	Dataflow Diagram
ER	Entity Relationship
CRUD	Create, Read, Update, and Delete

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Inventory management is a crucial aspect of business operations that ensures optimal stock levels, minimizes losses, and improves efficiency. Traditional manual inventory systems often lead to errors, inefficiency, and challenges in tracking product availability and sales. Inaccurate stock records may result in overstocking, understocking, or lost sales opportunities. Ineffective inventory management can also cause supply chain disruptions, increased operational costs, and customer dissatisfaction due to delays or unfulfilled orders.

With the rapid advancement of technology, businesses are shifting toward digital solutions for inventory management. A well-designed Online Inventory Management System can enhance the overall efficiency of businesses by automating key processes such as stock tracking, order management, and data analysis. Unlike traditional methods, a web-based system provides real-time insights into stock levels, transaction history, and business performance. This improves decision-making and ensures that inventory levels align with demand.

The primary goal of this project is to develop an Online Inventory Management System that simplifies inventory tracking, reduces human error, and enhances security measures. The system will offer a centralized platform where users can manage stock, monitor transactions, and generate reports with ease. Additionally, the system will provide role-based access control to ensure data security and prevent unauthorized modifications. By implementing this solution, businesses can significantly improve their inventory control, reduce operational costs, and maximize overall productivity.

This project contributes to the growing need for cost-effective and scalable inventory management solutions, especially for small and medium-sized businesses. By leveraging modern web technologies, the system ensures accessibility from any location with an internet connection, making it a convenient and reliable option for businesses seeking to improve their inventory operations.

1.2 Problem Statement

- Stock Mismanagement:** Inaccurate stock tracking results in overstocking or shortages, leading to financial losses.

- Traditional inventory management methods present several challenges: Inefficiency and Errors: Manual record-keeping is time-consuming and prone to human errors.
- Lack of Real-Time Updates: Delayed stock updates hinder effective business decision-making.
- Security Risks: Unauthorized access to inventory records can lead to data breaches and financial fraud.

This project aims to solve these issues by implementing an Online Inventory Management System that provides real-time stock updates, automated record-keeping, and secure authentication mechanisms.

1.3 Objectives

The main objectives of this project include:

1. To provide accurate, real-time inventory updates to prevent shortages and overstocking.
2. To implement authentication features to ensure only authorized users can access the system.
3. To provide sales, purchase, and stock level reports to support business decision-making.

1.4 Scope and Limitation

The scope and limitation of Online Inventory System (OIS) outlines the boundaries within which the learning experience and project contributions are defined.

1.4.1 Scope

1. Enables real-time tracking, updating, and organizing of stock levels, product details, and low-stock alerts to ensure smooth inventory flow.
2. Provides user authentication with different access levels (admin, users), along with accurate recording of sales and purchase transactions.
3. Generates and maintains essential records like sales summaries and inventory status, helping in organized documentation and quick reference.

1.4.2 Limitations

- The system is designed for small-scale businesses or individual use and may not support high-volume enterprise inventory needs.
- Its cannot be accessed in Mobile phones.
- Offline mode is not supported; requires continuous internet connectivity.

1.5 Report Organization

The report is organized into the following chapters:

Chapter 1: Introduction – Provides the project overview, problem statement, objectives, scope, and limitations.

Chapter 2: Background Study and Literature review – Details existing systems, Objectives, scope, and limitation.

Chapter 3: System analysis and Design – Covers requirement analysis, system design, and data modeling.

Chapter 4: Implementation and Testing -Discusses Test Cases.

Chapter 5: Conclusions and Future Recommendations – Summarizes the project Outcomes and suggest future enhancement.

CHAPTER 2: BACKGROUND AND LITERATURE REVIEW

2.1 Background Study

Inventory management systems play a crucial role in modern businesses by providing efficient ways to track stock levels, manage orders, and generate reports. Effective inventory management ensures that businesses maintain optimal stock levels, prevent stockouts, and minimize excess inventory.

Key concepts related to inventory management systems include:

- Stock Tracking:** This involves real-time monitoring of product quantities, ensuring that inventory levels are updated whenever stock is added, sold, or transferred.
- CRUD Operations:** Inventory management systems rely on four fundamental database operations: Creating, Reading, Updating, and Deleting (CRUD). These operations enable users to manage inventory records efficiently using MySQL databases.
- User Authentication:** Security is critical in inventory systems. Authentication mechanisms, implemented using PHP and MySQL, ensure that only authorized personnel can access sensitive data, reducing risks related to unauthorized modifications.
- Web-Based Systems:** This project utilizes a web-based approach, leveraging PHP, MySQL, HTML, CSS, and JavaScript to create an accessible platform that enables users to manage inventory remotely through a browser.
- Reporting and Analytics:** The system will generate automated reports using PHP and MySQL to help businesses make informed decisions regarding stock levels, sales trends, and reordering needs.

2.2 Literature Review

Existing Inventory Systems

Several inventory systems exist in the market, primarily designed for large enterprises with complex inventory needs. However, small and medium-sized businesses often struggle with

manual inventory tracking, leading to inefficiencies such as overstocking, understocking, and errors in record-keeping.

Examples of Existing Inventory Systems:

1.Zoho Inventory: A cloud-based inventory tracking and order management system designed for businesses of all sizes. It offers features such as multi-channel selling, order automation, and real-time stock tracking.

2.Trade Gecko (Now QuickBooks Commerce): A comprehensive inventory management platform tailored for small and medium businesses, integrating with e-commerce platforms and accounting software.

3.Odoo Inventory: An open-source inventory management solution with advanced features, including barcode scanning, automated replenishment, and warehouse management.

Need for an Affordable and Simple Inventory Management System

Despite the availability of these systems, small businesses often find them expensive, complex, and unsuitable for their specific needs. Many small enterprises still rely on spreadsheets or manual inventory tracking, leading to inefficiencies. The proposed Online Inventory Management System aims to bridge this gap by offering:

- Affordability: A cost-effective solution tailored for small to medium-sized businesses.
- Simplicity: A user-friendly interface developed using HTML, CSS, and JavaScript that requires minimal training.
- Essential Features: Stock tracking, user authentication, and automated reporting implemented using PHP and MySQL for seamless database interactions.
- Customization: A flexible system that can be adapted to different business needs, allowing businesses to modify inventory categories, manage user roles, and generate reports based on real-time data.

By utilizing PHP, MySQL, HTML, CSS, and JavaScript, this project will provide an efficient, scalable, and easy-to-use solution that enhances inventory management for small businesses.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

Methodology

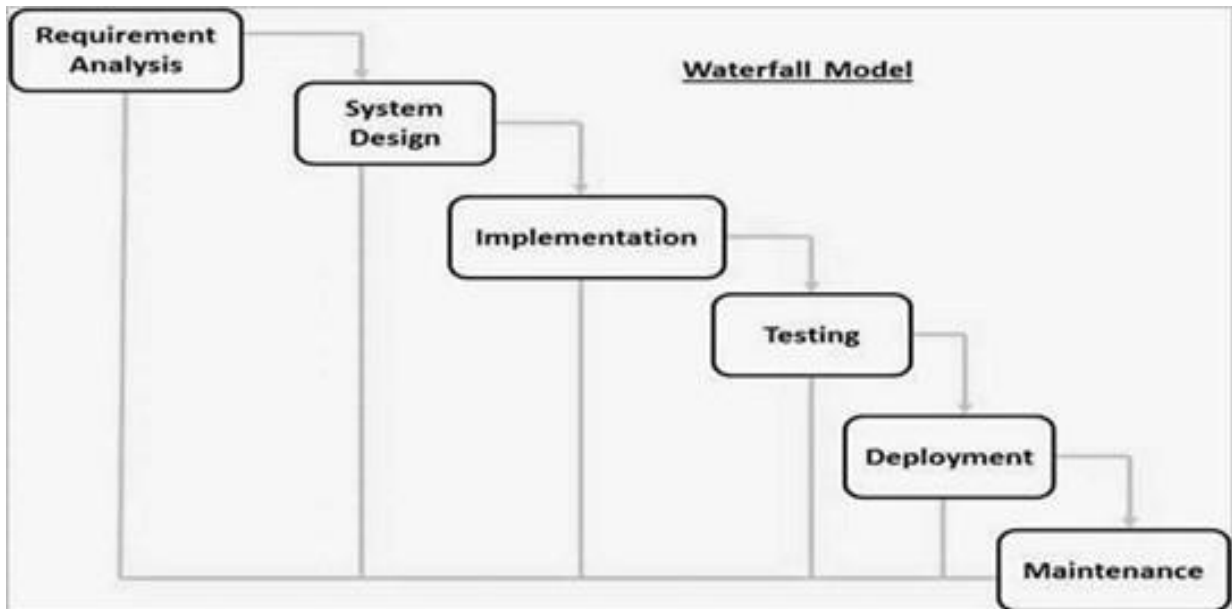


Fig 1: Waterfall model of OIS

Use Case of OIS

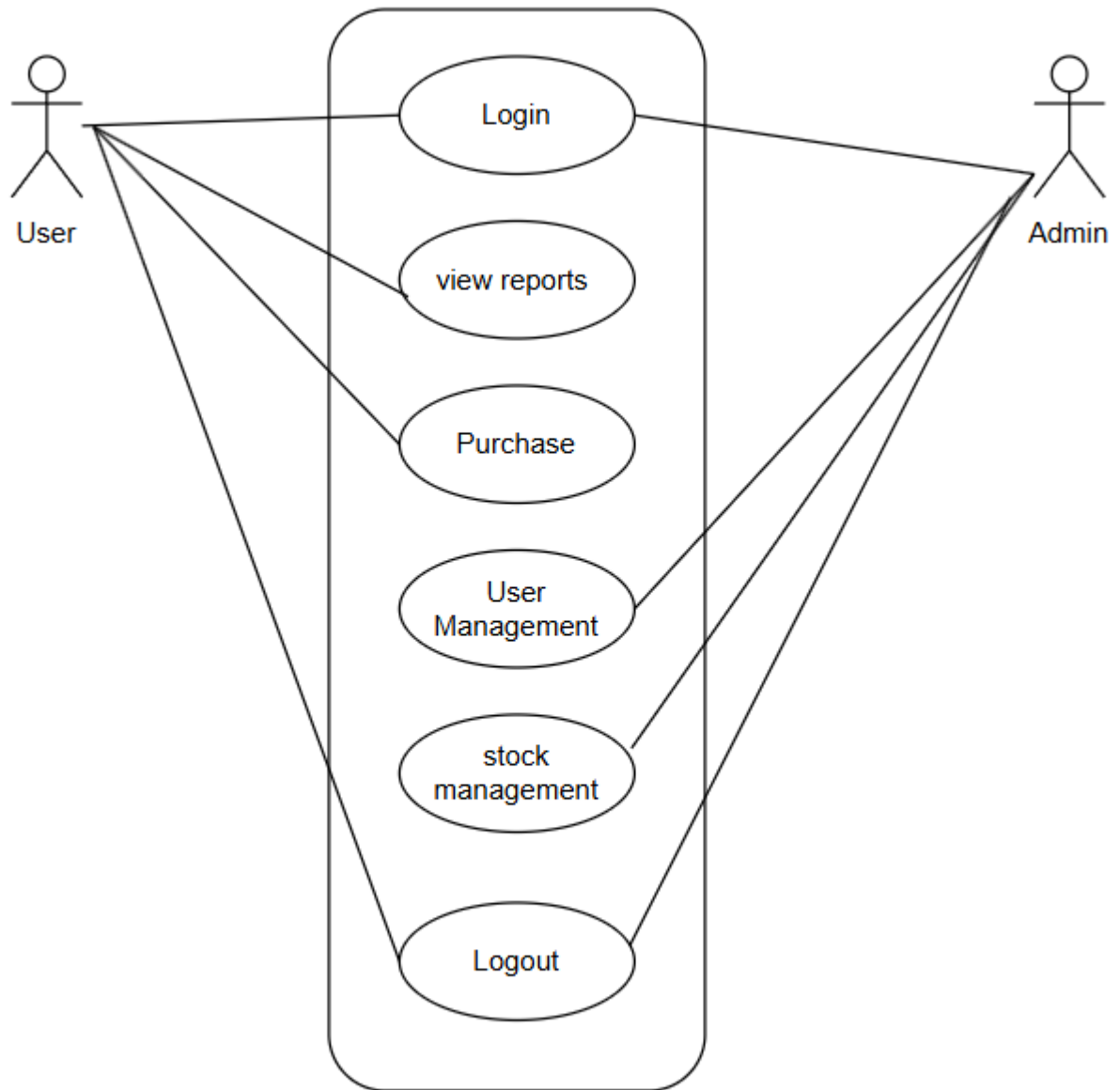


Fig 2: Use Case Diagram of OIS

3.1 Entity-Relationship (ER) Diagram of OIS

The image is an Entity-Relationship (ER) diagram for an inventory system, illustrating relationships between users, transactions, stock, and products. It includes attributes such as user details, product details, stock location, and transaction records.

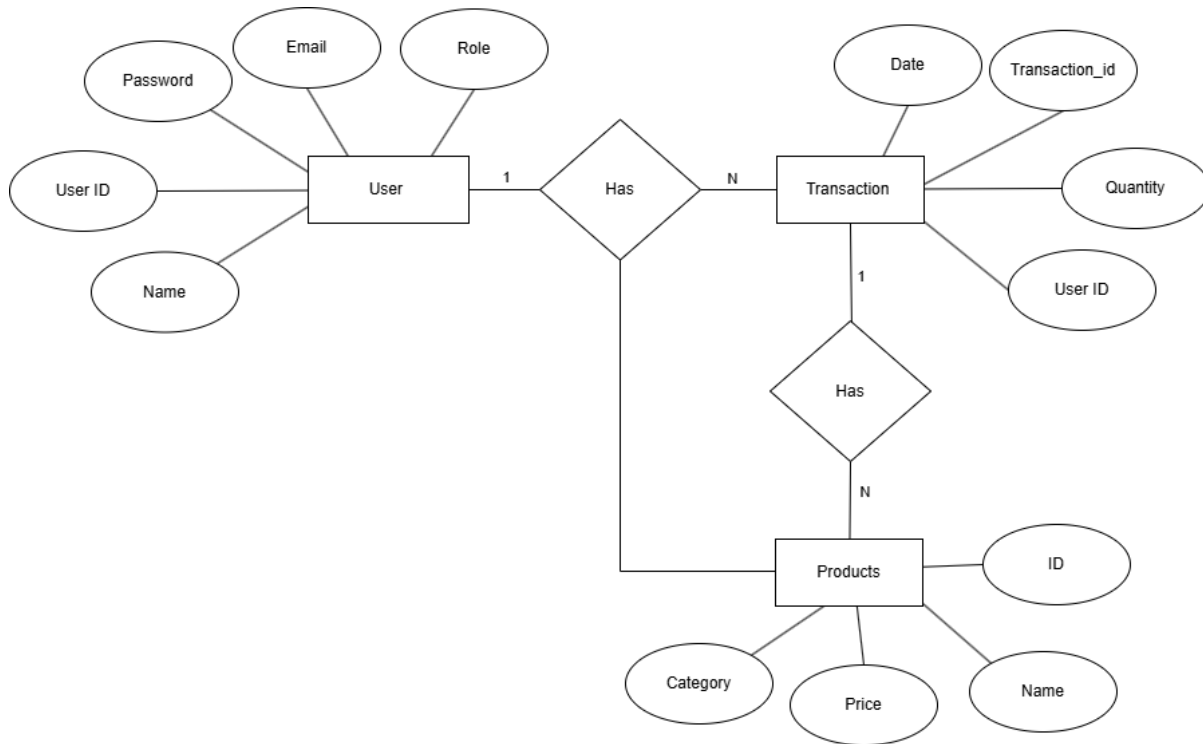


Fig 3: ER Diagram of OIS

3.2 Data Flow Diagram (DFD) of OIS

3.2.1 DFD Level 0

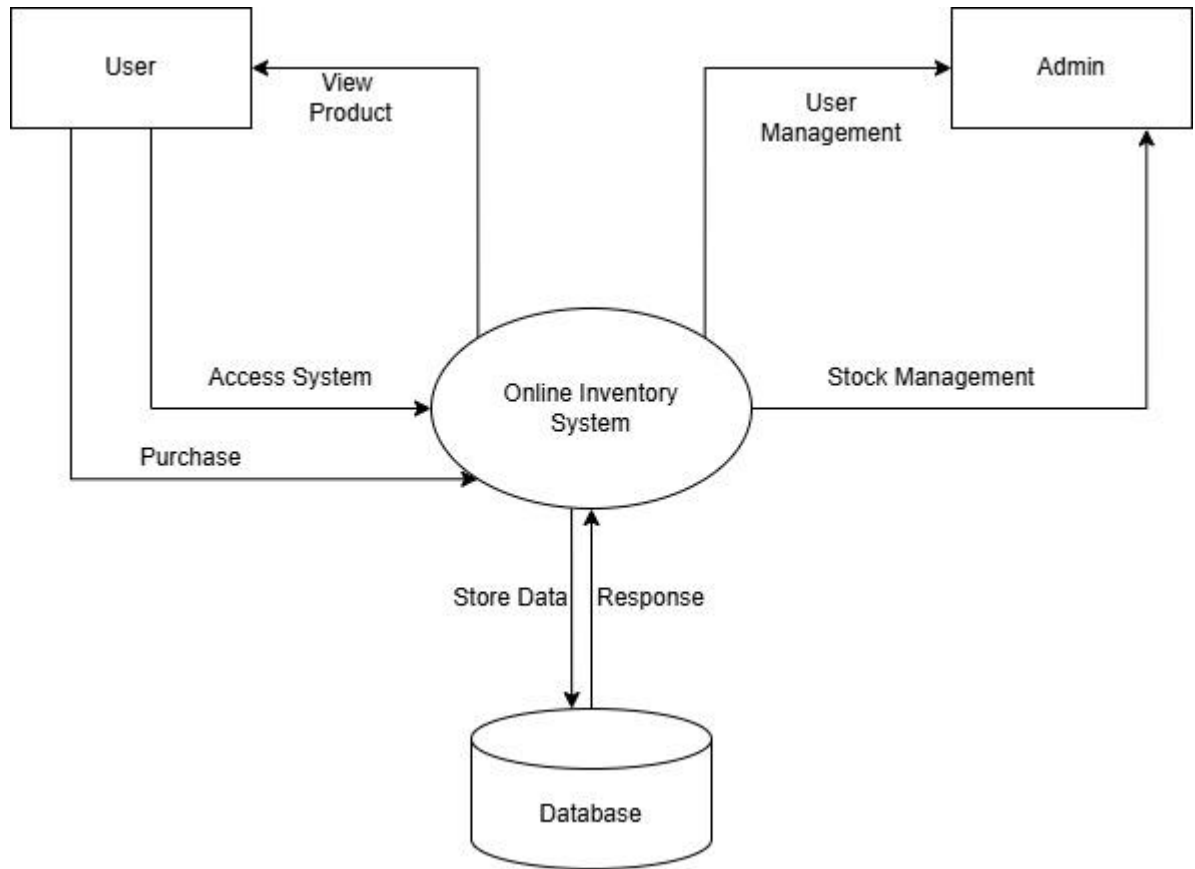


Fig 4: DFD Level 0 of OIS

The diagram represents an Inventory Management System where Admin/Users interact with modules like User Authentication, Product Management, Stock Management, and Transactions, all connected to a central database for storing product data. The system facilitates login, product tracking, stock updates, and transaction management.

3.2.2 DFD Level 1

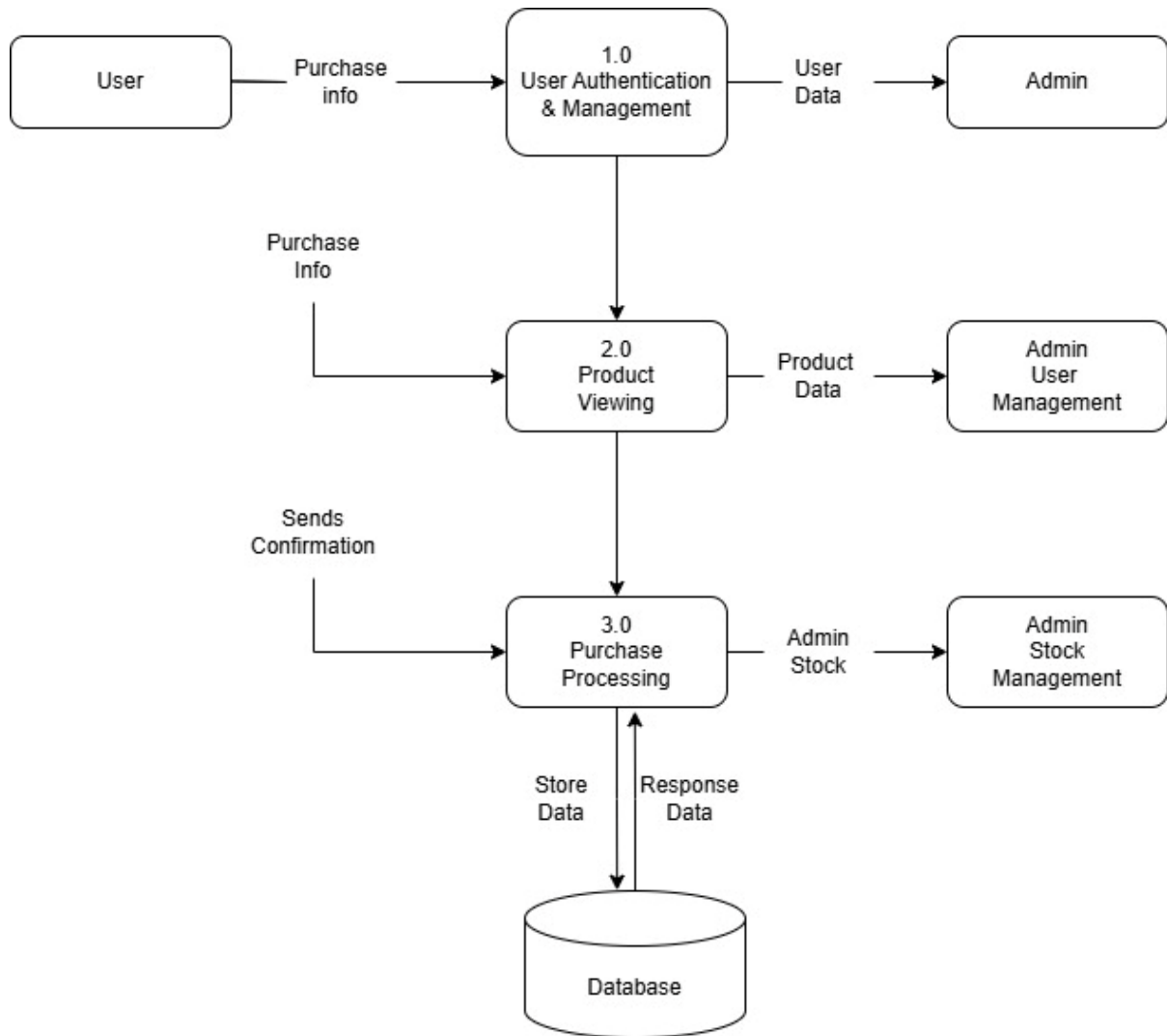


Fig 5: DFD Level 1 of OSI

The diagram outlines a three-module system: User Authentication (1.0), Product Viewing & Purchase Info (2.0), and Purchase Processing (3.0), all linked to a central Database. Admins manage user access, stock, and transactions, while users browse products and receive purchase confirmations. The flow ensures secure, real-time inventory tracking.

3.3 System Design

The system design phase includes:

3.3.1 Database design: Schema design using MySQL, defining collections for users, products and transactions.

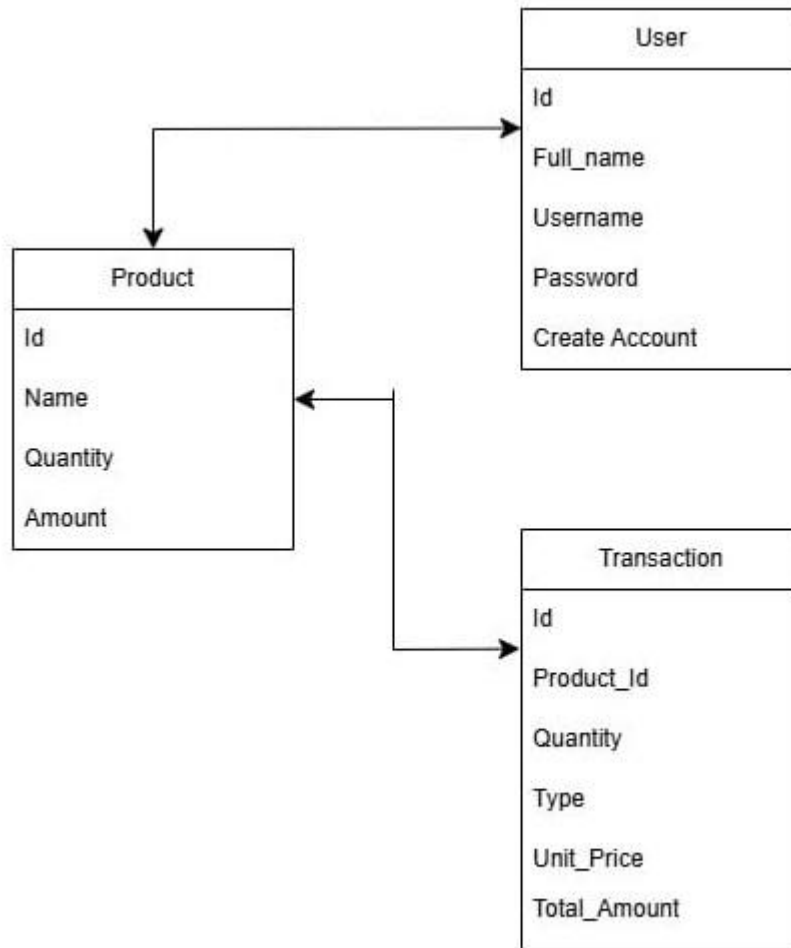


Fig 6: Database Design of OIS

3.3.2 System Architecture Diagram

The System Architecture Diagram illustrates the structure of an Inventory Management System by dividing it into five main layers:

1. User Interface (UI) – Includes pages such as Login, Register, Dashboard, Product Management, Transactions, and Stock Monitoring.
2. Backend – Handles system configuration, setup, and database interactions, processing requests from the UI.
3. Database Management – Stores and retrieves data using tables like Users, Products, Orders, and Transactions.
4. Authentication – Manages user login, registration, logout, and user-related operations.
5. External Services – Integrates with third-party services like Payment Gateways and Email Notifications.

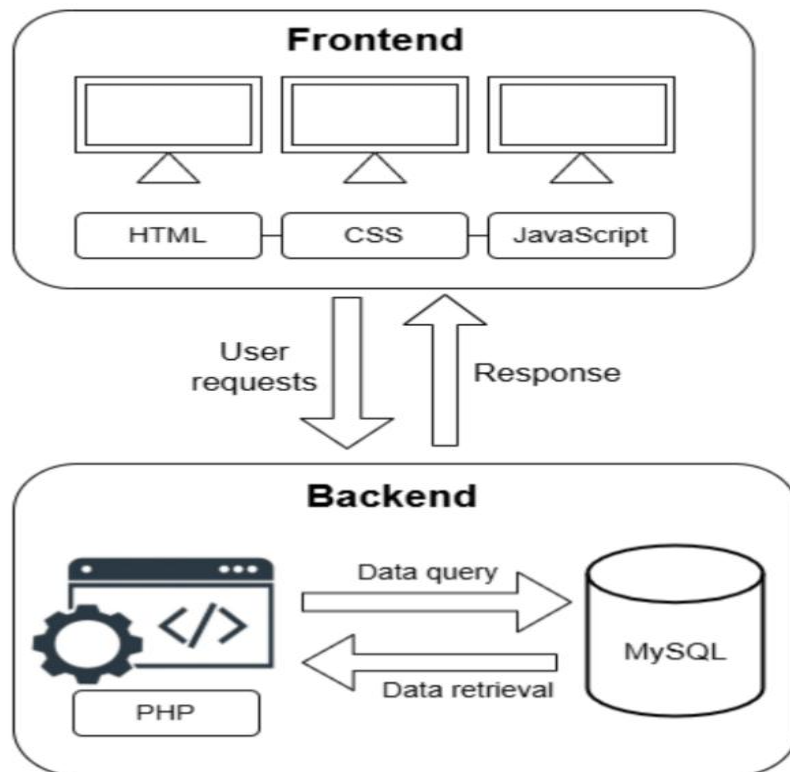


Fig 7: System Architecture Diagram of OIS

3.3.3 Block Diagram:

This block diagram represents the flow of actions and interactions within an Online Inventory System. Users start by signing up and creating an account, then proceed to log in using their credentials. After logging in, they can add, update, or remove inventory items and view reports about sales, stock status, and inventory trends.

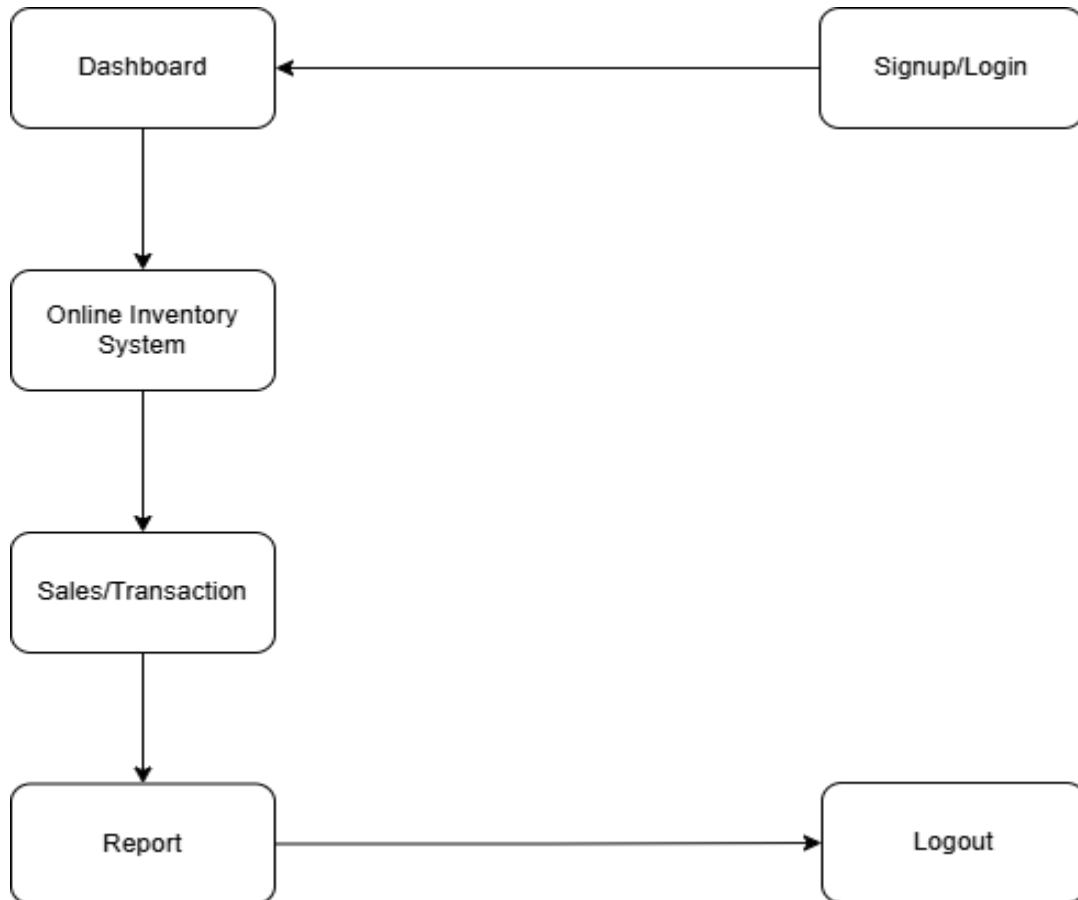


Fig 8: Block Diagram of OIS

3.4 Outcomes

Our project, after completion; is expected to look like template pictures shown below:

3.4.1 Login Page:

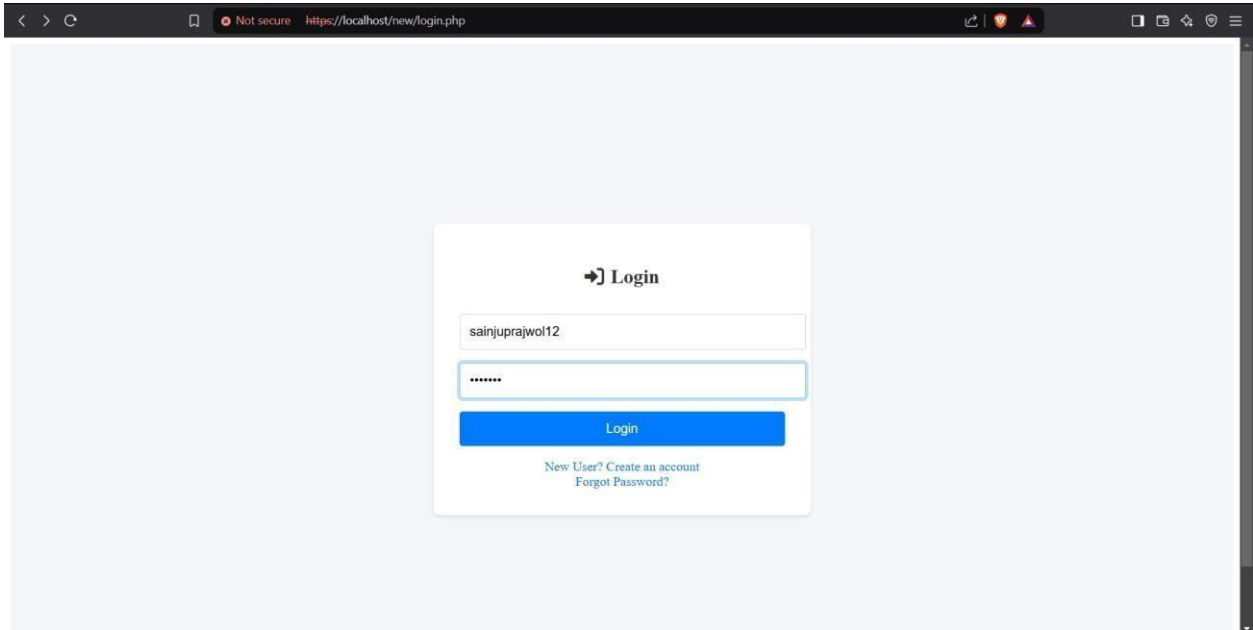


Fig 8: Login Page

3.4.2 Creating Account Interfaces:

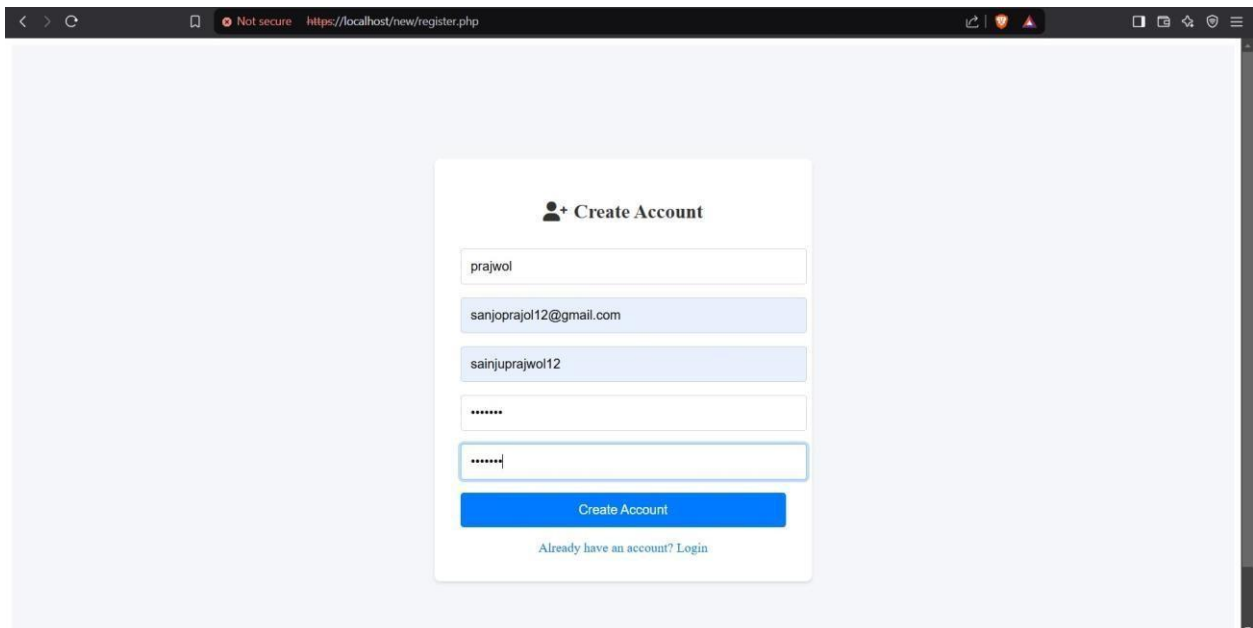


Fig 9: Creating Account Interfaces

3.4.3 Forgot Password Interfaces:

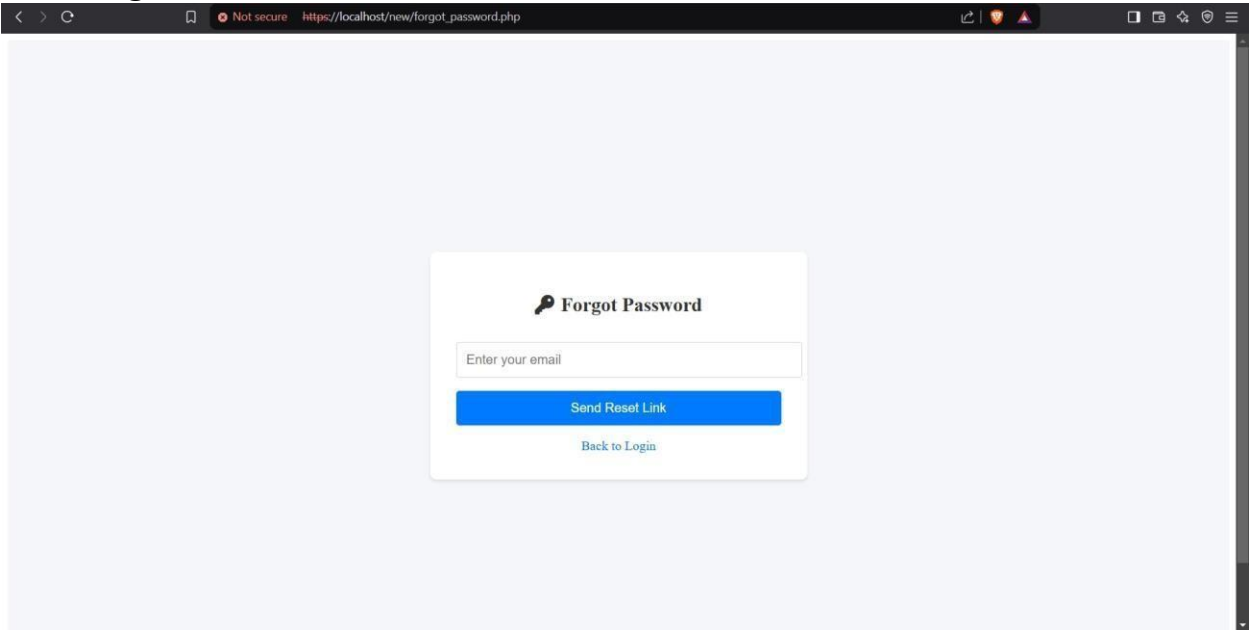


Fig 10: Forgot Password Interface

3.4.4 Home Pages:

3.4.4.1 Admin Interface

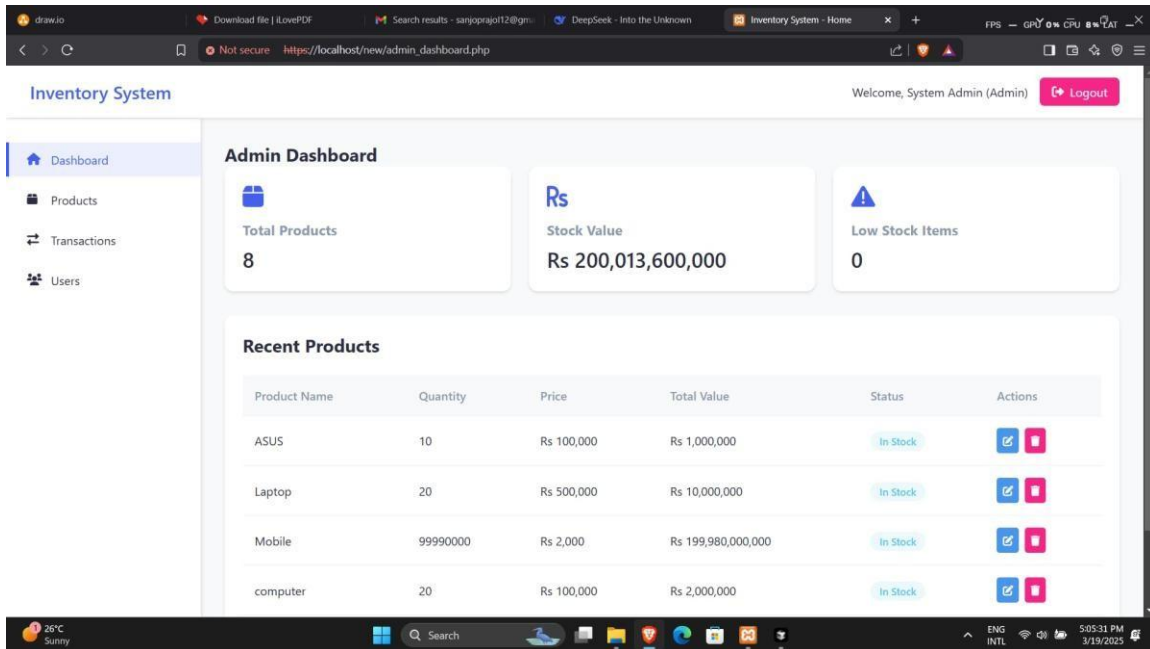


Fig 11: Admin Interface

3.4.4.2 User Interface

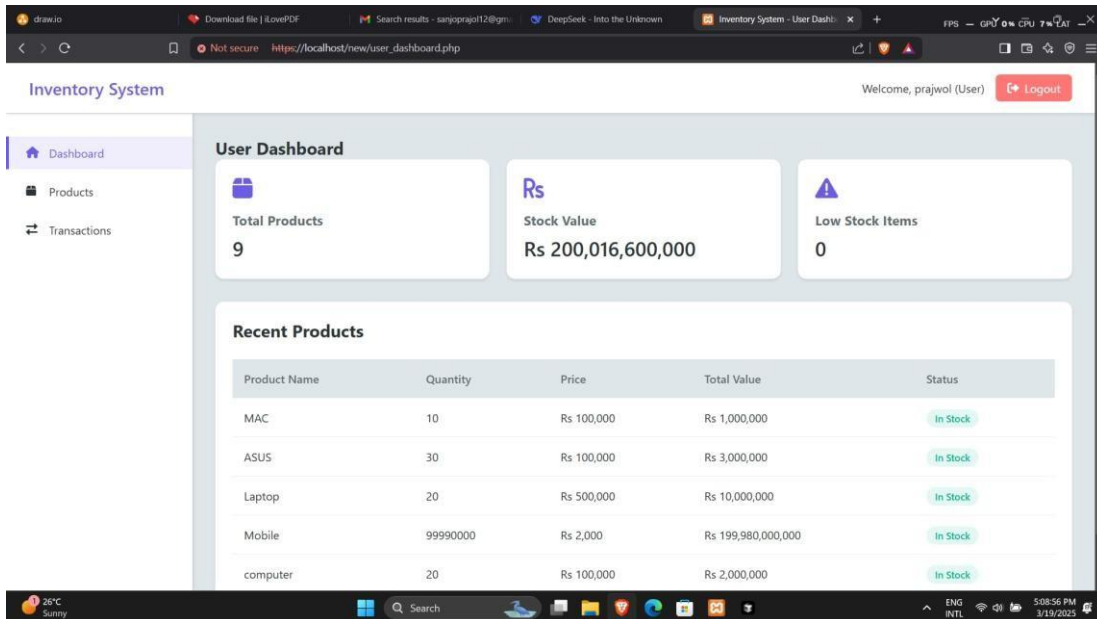


Fig 12: User Interface

3.4.5 Transactions:

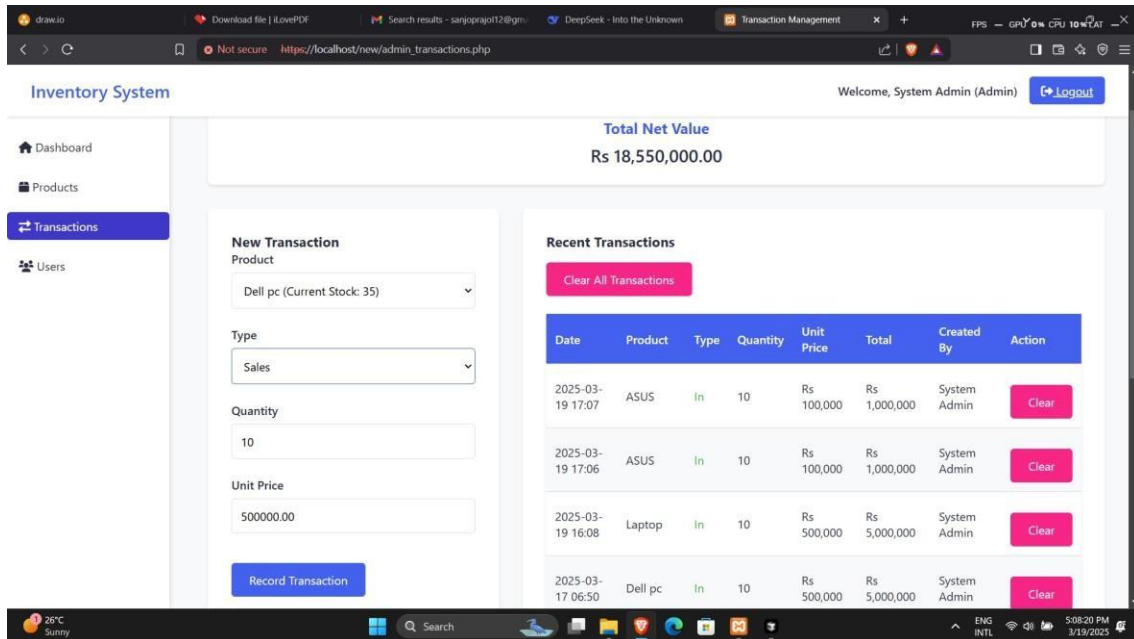


Fig 13: Transactions

3.4.6 Adding Products:

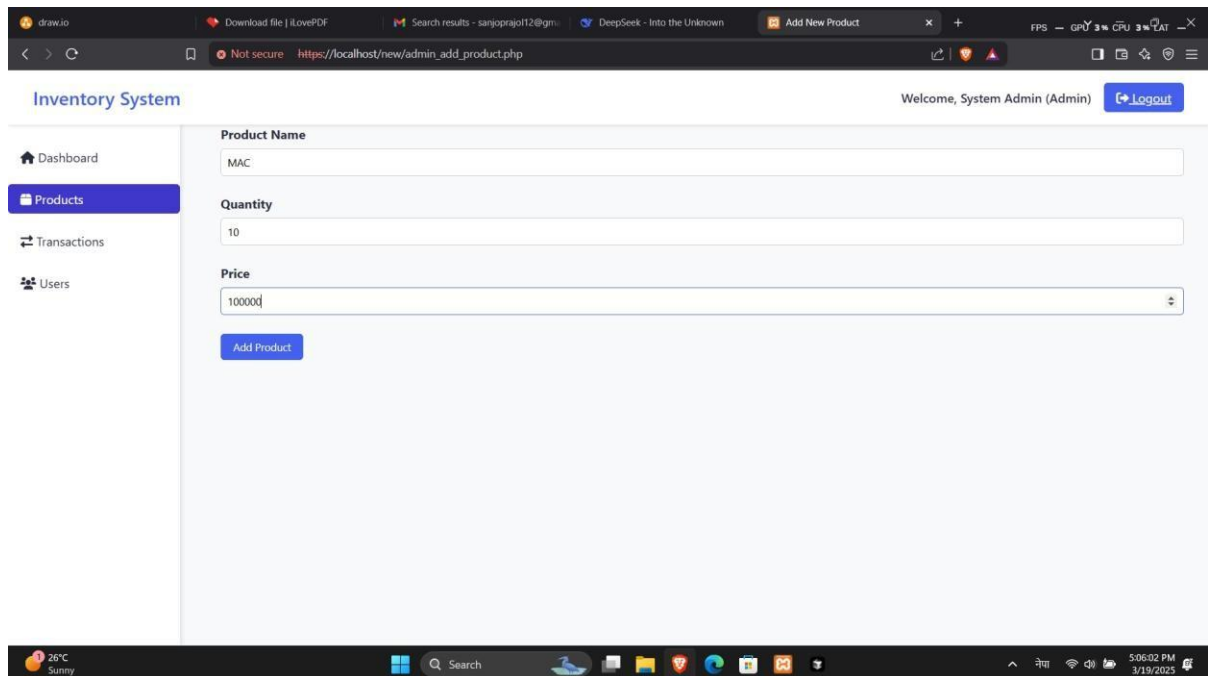
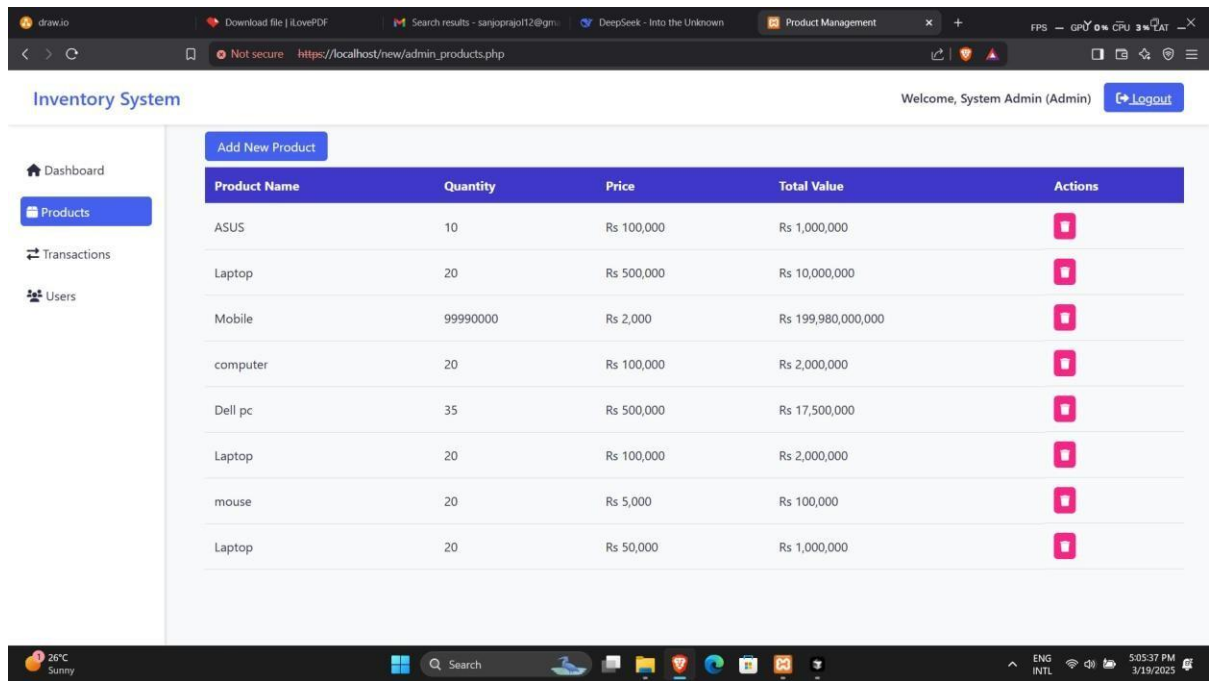


Fig 14: Adding Products

3.4.7Products:



The screenshot displays a web browser window with the URL `https://localhost/new/admin_products.php`. The page title is "Inventory System". In the top right corner, it says "Welcome, System Admin (Admin)" with a "Logout" button. On the left, there is a sidebar menu with "Dashboard", "Products" (highlighted), "Transactions", and "Users". The main content area features a table of products. Above the table is a button labeled "Add New Product". The table has five columns: "Product Name", "Quantity", "Price", "Total Value", and "Actions". Each row represents a product with its respective details and a delete icon in the "Actions" column.

Product Name	Quantity	Price	Total Value	Actions
ASUS	10	Rs 100,000	Rs 1,000,000	
Laptop	20	Rs 500,000	Rs 10,000,000	
Mobile	99990000	Rs 2,000	Rs 199,980,000,000	
computer	20	Rs 100,000	Rs 2,000,000	
Dell pc	35	Rs 500,000	Rs 17,500,000	
Laptop	20	Rs 100,000	Rs 2,000,000	
mouse	20	Rs 5,000	Rs 100,000	
Laptop	20	Rs 50,000	Rs 1,000,000	

Fig 15: Products

3.5 Gantt Chart

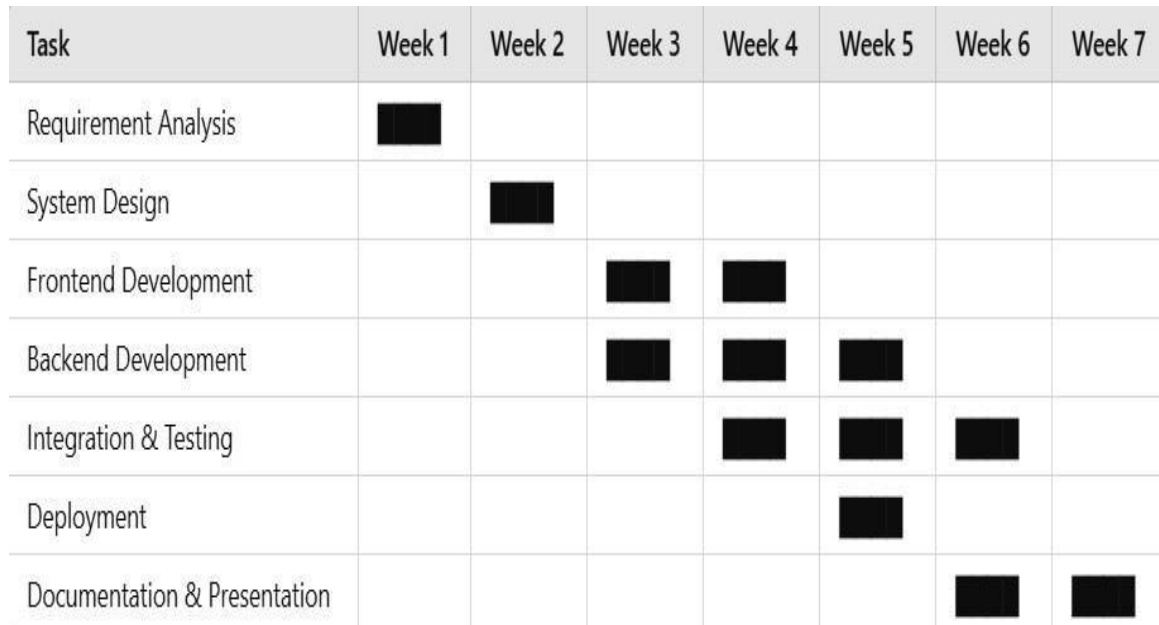


Fig 16: Gantt Chart

3.5 Time Schedule

Activities	Begin Date	End Date	Hours
Research	1st Sept 2024	2nd Sept 2024	12 hrs.
Project Planning	15th Sept 2024	25th Sept 2024	7 hrs.
Database Designing	5th Nov 2024	17th Nov 2024	11 hrs.
Coding	10th Dec 2024	16th Dec 2024	8 hrs.
Testing	21st Jan 2025	23rd Jan 2025	6 hrs.
Debugging	24th Jan 2025	2nd Feb 2025	8 hrs.
Final Project Submission	9th Feb 2025	19th Feb 2025	9 hrs.

Fig 17: Time Schedule

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

The implementation phase involves setting up the development environment, integrating various technologies, and ensuring that system functionalities work as expected. The system is developed using PHP, MySQL, HTML, CSS, JavaScript, and XAMPP for local hosting.

Tools Used

- PHP: Server-side scripting language for handling business logic and processing user requests.
- MySQL: Relational database management system for storing inventory and user data.
- HTML & CSS: Used for structuring and styling the web application.
- JavaScript: Enhances user interaction and real-time updates.
- XAMPP: Provides a local development environment, including Apache, MySQL, and PHP.

XAMPP for Local Hosting

To ensure efficient development and testing, XAMPP was used to set up a local server environment. It includes:

- Apache: Acts as the web server to host the application.
- MySQL: Manages the inventory database.
- PHP: Handles dynamic content and database interactions.

The system was developed and tested locally using XAMPP before deployment.

Implementation Details

1. User Authentication:

- Implemented using PHP and MySQL.
- Users can register, log in, and log out securely.
- Passwords are hashed for security.

2. Product Management:

- CRUD operations (Create, Read, Update, Delete) are implemented using PHP and MySQL.
- Users can add, modify, and delete product details.
- Each product entry includes fields like name, category, quantity, and price.

3. Stock Tracking:

- Real-time stock level updates using JavaScript and AJAX.
- When a product is added, updated, or sold, the database is immediately updated.

- Visual indicators alert users when stock levels are low.

4.Reporting System:

- PHP generates sales and inventory reports.
- Users can view daily, weekly, and monthly reports.
- Reports are exportable in PDF or CSV format.

4.2 Testing

During system testing, different training and testing datasets are used to verify whether the Online Inventory System (OIS) functions correctly with respect to user registration, product handling, and transaction management. The following unit tests were conducted to ensure accurate performance of each module:

4.2.1 Test Cases for Unit Testing

Table 4.1: Test case for Login and Inventory Operations

Title	Input	Expected Output	Test Result
Open Application	Open URL: http://localhost.login.php	Login Page	Pass
Registration Test	1. Full Name: Inventory User 2. User name: Invuser3 3. Email address: inventory12@gmail.com 4. Password: inv123 5. Confirm Password: inv123	Redirect To Login Page	Pass
Admin Login Test	1. Username: sainjuprajwol2 2. Password: sainju123	Redirect to the Admin Dashboard	Pass
User Login Test	1. Username: Invuser3 2. Password: inv123	Redirect to the User Dashboard	Pass
Invalid Login Test	Enter Wrong Username and Password 1.Username: ram 2. Password: ram123	Error: "Invalid Credentials"	Pass
Add Product Test	1. Go to Products. 2. Add Product Name: Mouse 3. Quantity: 10 4. Amount: 100	Product Added and Displayed in List.	Pass
Transaction Entry Test	1. Go to Transactions Page. 2. Select Product: Mouse 3. Select Type: Sales 4. Quantity: 9 5. Unit price: 100	Total amount: 900 Entry Added	Pass

Transaction View Test	1. Go to Transaction page	Beside that Transaction History Displayed Correctly	Pass
Admin Delete User Test	1. Select User: Invuser3 2. Click "Delete"	User Removed from System	Pass
Logout Test	1. Click Logout from Top Right corner from Dashboard.	Redirect To Login Page	Pass

CHAPTER 5: FUTURE RECOMMENDATION AND CONCLUSION

5.1 Conclusion

The **Online Inventory System** offers a modern, efficient, and secure solution to traditional inventory challenges by automating stock tracking, reducing errors, and enabling real-time updates. It enhances operational efficiency, prevents financial losses, and ensures data security through user authentication. Future upgrades could include mobile integration, AI-driven forecasting, and barcode/RFID support for better accuracy. Scalable and adaptable, this system is ideal for various industries, helping businesses optimize inventory management and boost productivity in the digital age.

5.2 Future Recommendations

To further improve the Online Inventory Management System, the following enhancements can be considered:

1. Mobile App Development – A mobile-friendly version to enable remote inventory management.
2. AI-Based Forecasting – Machine learning integration for demand prediction and stock optimization.
3. Blockchain Security – Enhanced data security and transparency in transactions.
4. RFID & Barcode Integration – Faster and more accurate stock tracking.
5. Cloud-Based Storage – Improved data accessibility and backup reliability.
6. Automated Supplier Management – Streamlined order tracking and supplier performance analysis.
7. E-commerce & ERP Integration – Synchronization with online stores and business management tools.

By implementing these recommendations, the system can become even more efficient, scalable, and valuable for businesses of all sizes.

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